

Stock Market Analysis in SQL: Insights Report

Prepared by Vipsa Patel
Data Analyst Intern, Labmentix

Summary

This report analyses six NSE stocks (Bajaj Auto, Eicher Motors, Hero MotoCorp, Infosys, TCS, and TVS Motors), 889 trading days each, from 1 Jan 2015 to 31 July 2018, using MySQL 8.0 and a Streamlit app. The course deck's Final Report says to sell TCS and Infosys because both look like big losers. That conclusion does not survive checking. Both companies issued 1-for-1 bonus shares (TCS on 31 May 2018, Infosys on 15 June 2015), which cut each share price roughly in half overnight without anyone losing value. After adjusting for these events, TCS moves from -23.8% to +52.4% and Infosys from -30.9% to +38.2%, and TCS's latest trend signal flips from sell to buy. The same price cliff also created false buy/sell signals in the moving average rule. The figures below comes from my own MySQL queries and the signal tables they produce, not from the deck. The one exception (Finding 8) is marked as computed in Python.

Price change over the full period

Stock	First close (1 Jan 2015)	Last close (31st July 2018)	Change, raw prices	Change, adjusted
TVS Motors	276.85	517.45	+86.9%	+86.9%
Eicher Motors	15,239.15	27,820.95	+82.6%	+82.6%
TCS	2,548.20	1,941.25	-23.8%	+52.4%
Infosys	1,975.80	1,365.00	-30.9%	+38.2%
Bajaj Auto	2,454.10	2,700.70	+10.0%	+10.0%
Hero MotoCorp	3,107.30	3,293.80	+6.0%	+6.0%

Source: Tasks 11 and 13. Adjusted = every price before the bonus date divided by 2. Only TCS and Infosys needed adjusting.

Finding 1: Price events must be checked before any ranking is trusted

Claim	TCS and Infosys each have a one-day price fall of about 50%, caused by a 1:1 bonus issue, not by a loss of value.
Evidence	Task 12 (worst single day per stock): TCS on 2018-05-31 fell 50.4% (close 1,744.80);Infosys on 2015-06-15 fell 49.9% (close 991.10). The other four stock s' worst days were between -6.2% and -9.7%. I confirmed both dates against public company and exchange announcements of 1:1 bonus issues (31 May 2018 for TCS, 15 June 2015 for Infosys).
Caveat	The adjustment divides prices by 2 and assumes no other corporate actions inside the data window. Dividends are not included. The two events are in different years, so the course deck's treatment of them as one problem is wrong too.

Finding 2: The deck's conclusion changes after adjustment

Claim	TCS and Infosys are the third- and fourth-best performers, not the two worst. The deck's recommendation to sell them is not supported.
Evidence	Task 11 vs Task 13 (table above). The deck also states Infosys fell 3%; my own query gives -30.9% on raw prices, so the deck figure is wrong even before adjustment. Smaller deck differences: Eicher is 82.6% in my query vs 82.5% in the deck (rounding) and the deck lists 2,454.1 as TCS's starting price, which is exactly Bajaj Auto's (TCS started at 2,548.20). Task 11b ranks the six by adjusted return with RANK(): TVS 1, Eicher 2, TCS 3, Infosys 4, Bajaj 5, Hero 6.
Caveat	Start-to-end change depends entirely on the two dates chosen. It says nothing about the path in between or about future returns.

Finding 3: The moving average signal rule fires rarely and the bonus cliff created false signals

Claim	A 20-day/50-day crossover rule produced 12 buy and 11 sell signals for Bajaj Auto across 889 days (866 hold). On raw prices TCS's latest signal (sell on 2018-06-05) was created by the price cliff.
Evidence	Tasks 5 to 10. Rule: Buy when the 20-day average crosses above the 50-day average, sell when it crosses below. Rebuilding TCS and Infosys signals on the adjusted prices (stretch query): TCS goes from 12 Buy/13 Sell with a last signal of Sell on 2018-06-05 to 12 Buy/12 Sell with a last signal of Buy on 2018-04-20. Infosys goes from 9/9 to 10/10, last signal Buy on 2018-05-07 (unchanged). Moving averages and signals for all 6 stocks are saved in the all_signals table (task 10B, 5,334 rows).
Caveat	Moving averages use only past prices, so the first possible signal needs 50 days of history and every signal arrives after the move has started. A signal is a trend flag not a forecast.

Buy/Sell signals, all six stocks (raw prices, Task 10)

Stock	Buys	Sells	Latest signal	Date	After correction
Bajaj Auto	12	11	Buy	2018-06-21	Unchanged
Eicher Motors	6	7	Sell	2018-06-06	Unchanged
Hero MotoCorp	9	9	Sell	2018-05-22	Unchanged
Infosys	9	9	Buy	2018-05-07	10/10, still buy
TCS	12	13	Sell	2018-06-05	12/12, Buy (2018-04-20)
TVS Motors	8	8	Sell	2018-05-17	Unchanged

After correction the latest signals are Buy for Bajaj Auto, Infosys and TCS, and Sell for Eicher, Hero, and TVS. The deck's list (Buy Bajaj and Infosys; Sell Eicher, TVS, Hero, and TCS) changes in one place: TCS moves from Sell to Buy.

Finding 4: Signals can flip back and forth quickly and that is costly

Claim	In Bajaj Auto's signal history, quick reversals cost money: 6 of its 11 completed buy-then-sell round trips lasted 30 trading days or less, and 5 of those 6 lost money.
Evidence	From the bajaj2 table (query: SELECT date, close_price, 'signal' FROM bajaj2 WHERE 'signal' <> 'Hold'). Example: a buy on 2015-12-28, followed by a sell on 2016-01-18 (15 trading days) lost about 9.6% if traded at those closing prices. Another buy on 2018-02-01 was reversed by a sell on 2018-02-06 (3 trading days) for about -8.0%.
Caveat	Trades are assumed at the closing price on the signal day, with no brokerage, taxes or slippage, so real losses would be larger. This analysis covers Bajaj Auto only and was done on the signal table, not a separate query per stock.

Finding 5: The latest signal and the long-run trend can disagree

Claim	TVS Motors (+86.9%) and Eicher Motors (+82.6%) are the two strongest long-run performers yet both ended on a sell signal.
Evidence	Task 10 (latest signals: TVS sell 2018-05-17, Eicher sell 2018-06-06) against Task 11 (price change).
Caveat	The signal reacts to the last 20 to 50 days; the long-run change is measured with hindsight. I would use the long-run change to judge how a stock has performed and treat the signal only as a short-term momentum flag. Neither predicts the future.

Finding 6: Yearly averages from raw prices can mislead

Claim	TCS's yearly average close on raw prices: (2,537.39, 2,419.00, 2,475.36, and 2,729.12 for 2015 to 2018, Task 3) cannot be read as growth. On the bonus-adjusted prices the averages are 1,268.69, 1,209.50, 1,237.68, and 1,646.45 (Task 3).
Evidence	Task 3 (raw) and Task 3b (adjusted) on the tcs table.
Caveat	2018 covers only 7 months. The raw 2018 average mixes pre-bonus prices (January to May) with post-bonus prices (June and July), so it is not comparable with earlier years. Task 3b divides the pre-bonus prices by 2 to fix this.

Finding 7: The data is nearly complete

Claim	All six tables have 889 rows from 2015-01-01 to 2018-07-31. The only gaps are in the deliverable quantity column, on two dates.
Evidence	Tasks 1 and 4: 2015-12-09 is missing for Eicher, Hero, TCS and TVS; 2017-08-31 is missing for Bajaj and Infosys.
Caveat	I checked only that column for NULLs. The same two dates appearing across unrelated companies suggests an exchange reporting gap, but I have not confirmed the cause. The delivered quantity column is not used in any signal.

Finding 8: Following the signals did not beat simply holding

Claim	Following the 20-day/50-day signals would not have beaten buy-and-hold for any of the six stocks even before costs.
Evidence	Strategy vs Hold tab of the Streamlit app (computed in Python, not SQL). Rule: buy at the close on each Buy signal, sell at the close on each Sell signal, stay in cash in between, on bonus-adjusted prices, with any open trade valued at the last close. Total return 2015 to 2018 is in the table below. Live app: https://stocksense-nse.streamlit.app/
Caveat	One rule, six stocks and about 3.5 years, so this does not prove that signals never work. Trades are assumed at the closing price on the signal day. Costs are shown at 0.5% per buy and per sell; taxes, slippage, and dividends are not modelled.

Stock	Buy and hold	Follow the signals	Signals, 0.5% cost per trade	Trades
TVS Motors	+86.9%	+85.0%	+70.7%	8
Eicher Motors	+82.6%	+76.5%	+66.1%	6
TCS	+52.4%	-22.4%	-30.9%	12
Infosys	+38.2%	+7.1%	-2.6%	10
Bajaj Auto	+10.0%	-2.7%	-13.4%	12
Hero MotoCorp	+6.0%	-16.3%	-23.6%	9

Conclusions

- **Check for corporate actions first.** Bonus issues created the two biggest apparent losses in this data; comparing raw prices across them gives the wrong answer.
- **The deck's sell call for TCS and Infosys is not supported.** On corrected prices they are the third and fourth best performers, and TCS's latest signal is Buy.
- **A 20/50-day crossover is a trend indicator, not a winning strategy here.** It did not beat buy and hold on any of these six stocks.

Method limits: what this data cannot tell an investor

- **Lagging signals:** Moving averages use only past prices, so a crossover confirms a move that has already happened.
- **Missing information:** dividends, company news, brokerage and tax costs, and the broader market are not in this data.
- **Small sample:** about 3.5 years of data, with 6 stocks from 2 sectors (two-wheelers and IT services).
- **Not investment advice:** this is a SQL exercise in finding and correcting a data trap.